Problem H1: Lorentz-Invariant Differential Cross Section

Converting p_i^* to Mandelstam Variables & Feynman Diagrams in the ABC Theory

Based on Griffiths Ch. 6 / Thomson §3.5

Particle Physics Study Guide

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1 Problem Statement (Verbatim)

Problem Statement

H1) Using the expression for $d\sigma/dt$ for $1+2 \to 3+4$ (natural units throughout). Given from Thomson Book:

(a) Express the formula using only Mandelstam variables if $m_1 = m_2 = m$ (& other constants but for all kinematic factors... i.e. instead of $|\mathbf{p}_i^*|$ or $|\mathbf{p}_f^*|$).

(b) [Somewhat like Griffiths 6.13 or 6.14 but for any frame] Use it and the Feynman rules for the ABC theory to derive the cross section $d\sigma/dt$ for $C+C \to A+A$ scattering to lowest order, also assuming propagating particle mass of 0, and the other two particle types have the same non-zero mass m. Write the final expression in terms of only Mandelstam variables ($= f(s,t,u)$). You may follow the similar short derivation in the book [note, for slightly different particles used], or go from commentary on that in the slides, but at least start by writing the diagrams and at least briefly justify why the ABC Feynman rules mean results wouldn't change essentially.

2 Reworded Goals, Objectives, and What It Wants

Part (a): The Goal

You are given the Lorentz-invariant differential cross section formula from Thomson Eq. 3.37:

$$\frac{d\sigma}{dt} = \frac{1}{64\pi\, s\, p_i^{*2}} |\mathcal{M}|^2$$

This formula still contains p_i^{*2}—the CM-frame initial momentum squared. Your job:

1. Express p_i^{*2} purely in terms of s and the common mass m (using $m_1 = m_2 = m$).
2. Write the final formula so that *every kinematic quantity* is a Mandelstam invariant or a mass parameter—nothing that "looks" frame-dependent.

Why: This gives a universal formula you can plug *any* $|\mathcal{M}|^2$ into and get $d\sigma/dt$ in *any* frame.

Part (b): The Goal

Apply the result of (a) to a specific process in the ABC toy theory (Griffiths §6.3):

1. **Draw all lowest-order Feynman diagrams** for $C + C \to A + A$—and explain *why* those are the right diagrams (which channels exist, which don't, and why).
2. **Apply the Feynman rules** step-by-step to extract the amplitude $\mathcal{M}$ from each diagram.
3. **Add the amplitudes** coherently and square.
4. **Plug into Part (a)** to get $d\sigma/dt$ purely in terms of s, t, u and constants (g, m).
5. **Justify** why the answer has the same structure as Griffiths' $A+A \to B+B$ calculation (Eqs. 6.54–6.59)—the vertex structure is identical, so only mass labels change.

3 Part (a): Converting p_i^{*2} to Mandelstam Variables

3.1 The Starting Formula

Thomson Eq. 3.37: Lorentz-Invariant $d\sigma/dt$

For any $2 \to 2$ scattering process $1 + 2 \to 3 + 4$:

$$\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{64\pi\, s\, p_i^{*2}} \tag{1}$$

where $p_i^* \equiv |\mathbf{p}_1^*| = |\mathbf{p}_2^*|$ is the magnitude of either initial-state 3-momentum in the CM frame, and $s = (p_1 + p_2)^2$.

Where does this come from? Briefly: start from Griffiths' CM cross section formula (Eq. 6.47 / Appendix B Eq. B.6):

$$\frac{d\sigma}{d\Omega^*} = \frac{S}{64\pi^2} \frac{|\mathbf{p}_f^*|}{E_{\text{CM}}^2\, |\mathbf{p}_i^*|} |\mathcal{M}|^2$$

Then change variables from $d\Omega^* = d(\cos\theta^*)\, d\phi^*$ to dt using the relation $dt = 2\, p_i^*\, p_f^*\, d(\cos\theta^*)$, integrate over ϕ^* (picking up 2π), and use $E_{\text{CM}}^2 = s$. The factors of $|\mathbf{p}_f^*|$ cancel, and you arrive at Eq. (1). **Note:** In this Lorentz-invariant form, the statistical factor S is absorbed into how you define the limits of integration over t, or can be carried explicitly. We keep S explicit throughout.

3.2 Step 1: p_i^{*2} in Terms of s and Masses (General)

General CM Momentum (Thomson Eq. 3.38)

$$p_i^{*2} = \frac{\left[s - (m_1 + m_2)^2\right]\left[s - (m_1 - m_2)^2\right]}{4s} \tag{2}$$

Full derivation (since the problem is really about showing this step):

In the CM frame: $\mathbf{p}_1^* = -\mathbf{p}_2^*$, both with magnitude p_i^*. Define $\sqrt{s} = E_1^* + E_2^*$. Each particle is on-shell:

$$E_1^{*2} = p_i^{*2} + m_1^2, \qquad E_2^{*2} = p_i^{*2} + m_2^2$$

Write $E_2^* = \sqrt{s} - E_1^*$ and square:

$$p_i^{*2} + m_2^2 = s - 2\sqrt{s}\, E_1^* + p_i^{*2} + m_1^2$$

The p_i^{*2} cancels. Solve:

$$E_1^* = \frac{s + m_1^2 - m_2^2}{2\sqrt{s}} \tag{3}$$

Then $p_i^{*2} = E_1^{*2} - m_1^2$:

$$p_i^{*2} = \frac{(s + m_1^2 - m_2^2)^2}{4s} - m_1^2 = \frac{(s + m_1^2 - m_2^2)^2 - 4s\, m_1^2}{4s} \tag{4}$$

Expand the numerator:

$$s^2 + 2s(m_1^2 - m_2^2) + (m_1^2 - m_2^2)^2 - 4s\, m_1^2 = s^2 - 2s(m_1^2 + m_2^2) + (m_1^2 - m_2^2)^2$$

This factors as $[s - (m_1 + m_2)^2][s - (m_1 - m_2)^2]$, giving Eq. (2).

3.3 Step 2: Specialize to $m_1 = m_2 = m$

$$p_i^{*2} = \frac{[s - 4m^2][s - 0]}{4s} = \frac{s\,(s - 4m^2)}{4s}$$

Result: CM Momentum for Equal Masses

$$\boxed{p_i^{*2} = \frac{s - 4m^2}{4}} \tag{5}$$

3.4 Step 3: The Final Lorentz-Invariant Formula

Substitute Eq. (5) into Eq. (1):

$$\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{64\pi\, s \cdot \dfrac{s - 4m^2}{4}} = \frac{4\,|\mathcal{M}|^2}{64\pi\, s\,(s - 4m^2)}$$

Part (a) Final Answer

$$\boxed{\frac{d\sigma}{dt} = \frac{|\mathcal{M}|^2}{16\pi\, s\,(s - 4m^2)}} \tag{6}$$

This is valid for **any** $2 \to 2$ process with $m_1 = m_2 = m$, in **any** reference frame. Multiply by S when computing total cross sections (or include it here and use the full integration range for t).

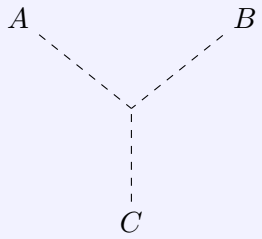
4 Part (b): The Feynman Diagrams for $C + C \to A + A$

This is the heart of the problem. We need to carefully identify the correct Feynman diagrams before we can compute anything.

4.1 The ABC Theory: What You're Allowed to Draw

The ABC Vertex Rule (Griffiths §6.3)

The ABC theory has **exactly one primitive vertex**. It connects **one** A **line**, **one** B **line**, and **one** C **line**:



Vertex factor: $-ig$

Every vertex in **every** diagram must have exactly one A, one B, one C. No exceptions. There is no AAB vertex, no CCA vertex, no BBB vertex, etc.

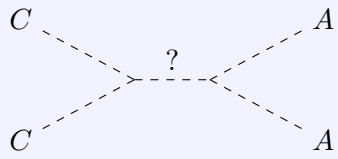
4.2 How to Decide Which Diagrams to Draw

Systematic Diagram Identification

For $C(p_1) + C(p_2) \to A(p_3) + A(p_4)$ at lowest order (two vertices, one internal line), we ask: **which particle can the internal line be?**

At each vertex, we need one A, one B, one C. Consider each channel:

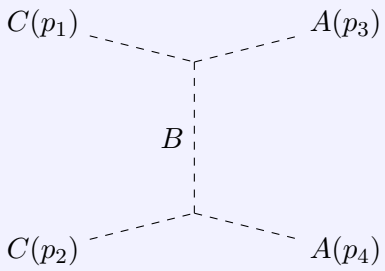
(i) Can there be an s-channel diagram?
An s-channel diagram would look like: both incoming C's meet at a vertex, annihilate into a virtual particle, which then produces both outgoing A's at a second vertex.



FORBIDDEN

The left vertex has two C's and one internal line. For a legal ABC vertex, we'd need one A, one B, one C—but we have **two C's.** $\Longrightarrow$ **No s-channel diagram exists.**

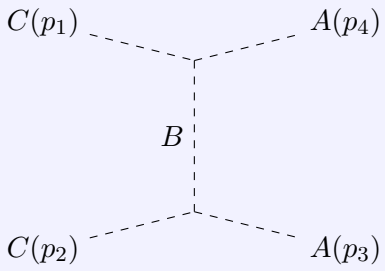
(ii) Can there be a t-channel diagram?
Route $C(p_1)$ to $A(p_3)$ at one vertex, and $C(p_2)$ to $A(p_4)$ at the other:



ALLOWED

Top vertex: one C, one A, one B (the internal line). ✓
Bottom vertex: one C, one A, one B. ✓
$\Longrightarrow$ **The t-channel diagram is legal.** The virtual particle is a B.

(iii) Can there be a u-channel diagram?
This is the "twisted" version: route $C(p_1)$ to $A(p_4)$ and $C(p_2)$ to $A(p_3)$:



ALLOWED

Both vertices again have one A, one B, one C. ✓
$\Longrightarrow$ **The u-channel diagram is also legal.**

Conclusion: There are exactly **two** lowest-order diagrams: t-channel and u-channel exchange of a virtual B.

Why This Is "Just Like" Griffiths' $A + A \to B + B$

In Griffiths §6.3.2 (Eqs. 6.54–6.59), the process $A + A \to B + B$ is worked out. That process also has:

- **No s-channel**: two A's can't meet at a single ABC vertex (you'd need A-A-something, which violates the one-of-each rule).
- **Two diagrams**: t-channel and u-channel, with internal line = C.
- **Identical topology**: The "twisted" diagram (Griffiths Fig. 6.9) is obtained by swapping the two identical outgoing particles, $p_3 \leftrightarrow p_4$.

The **only** difference between Griffiths' problem and ours is which particle is on the external lines and which is on the internal line:

	Griffiths ($A + A \to B + B$)	Our problem ($C + C \to A + A$)
Incoming	A, A	C, C
Outgoing	B, B	A, A
Internal	C (mass m_C)	B (mass $m_B = 0$)

Since the ABC vertex is *symmetric*—it doesn't care which of A, B, C is incoming vs. outgoing vs. internal—the vertex factor is $-ig$ in *every* case. The algebraic structure of $\mathcal{M}$ is **identical**. Only the mass in the propagator denominator changes.

5 Part (b): Applying the Feynman Rules Step-by-Step

5.1 The ABC Feynman Rules (Griffiths §6.3)

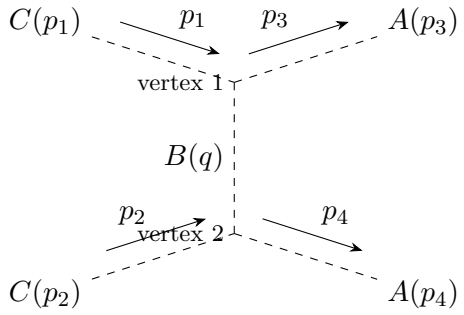
Griffiths' Feynman Rules for the ABC Theory

To compute $-i\mathcal{M}$ for a given diagram:

1. **Notation:** Label all external four-momenta $p_1, p_2, \ldots$ (arrows forward in time). Label internal momenta $q_1, q_2, \ldots$ (arrows arbitrary).
2. **Vertex factor:** For each vertex, write $-\mathbf{ig}$.
3. **Propagator:** For each internal line carrying four-momentum q_j and mass m_j, write
$$\frac{i}{q_j^2 - m_j^2}$$
(The virtual particle is *off-shell*: $q_j^2 \neq m_j^2$ in general.)
4. **Momentum conservation:** Impose 4-momentum conservation at each vertex via a delta function $(2\pi)^4\,\delta^{(4)}(\sum k)$.
5. **Integration:** For each internal line, integrate $\displaystyle\int \frac{d^4q}{(2\pi)^4}$.
6. **Cancel the overall $\delta^{(4)}$:** Erase the final $(2\pi)^4\,\delta^{(4)}(p_1+p_2-p_3-p_4)$ and multiply by i. The result is $\mathcal{M}$.

Note: For tree-level diagrams (no loops), the delta functions at each vertex fix all internal momenta, so no integrals remain. This is why tree-level calculations are straightforward.

5.2 Diagram 1: The t-channel



Apply the rules line by line:

Step 1 – Vertices: Two vertices $\Rightarrow$ two factors of $(-ig)$.
$$(-ig) \times (-ig) = (-ig)^2 = -g^2$$

Step 2 – Propagator: One internal B line with mass m_B:
$$\frac{i}{q^2 - m_B^2}$$

Step 3 – Momentum conservation: At vertex 1: $p_1 = p_3 + q$, so $q = p_1 - p_3$. Therefore:

$$q^2 = (p_1 - p_3)^2 \equiv t$$

Step 4 – Assemble: (After performing the trivial integral and canceling the overall $\delta^{(4)}$):

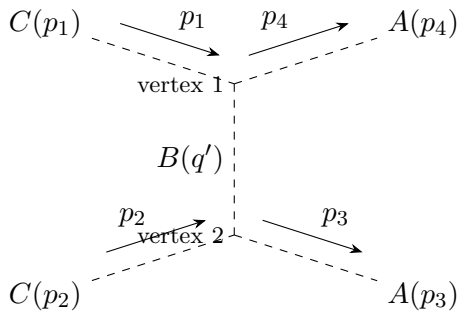
$$-i\mathcal{M}_t = (-ig)^2 \cdot \frac{i}{t - m_B^2}$$

Simplify: $(-ig)^2 = -g^2$, so $(-g^2)(i) = -ig^2$:

$$\boxed{-i\mathcal{M}_t = \frac{-ig^2}{t - m_B^2}} \tag{7}$$

5.3 Diagram 2: The u-channel (the "twisted" diagram)

This is Griffiths' key observation (Fig. 6.9): the second diagram is obtained by **swapping the two outgoing A particles**, i.e., $p_3 \leftrightarrow p_4$.



The **only** change: at vertex 1, momentum conservation now gives $q' = p_1 - p_4$, so:

$$q'^2 = (p_1 - p_4)^2 \equiv u$$

Since the vertex factors and propagator structure are identical, we can write the answer directly by replacing $t \to u$ in Eq. (7):

$$\boxed{-i\mathcal{M}_u = \frac{-ig^2}{u - m_B^2}} \tag{8}$$

The "Twisting" Trick (Griffiths §6.3.2)

Griffiths emphasizes: *"Since this differs from [the first diagram] only by the interchange $p_3 \leftrightarrow p_4$, there is no need to compute it from scratch."* This is a general principle: when you have identical particles in the final state, the second diagram is always obtained by swapping their momenta, which sends $t \leftrightarrow u$.

5.4 Step 5: Total Amplitude

Both diagrams contribute to the *same* physical process, so we **add amplitudes before squaring** (quantum mechanics!):

$$-i\mathcal{M} = -i\mathcal{M}_t + (-i\mathcal{M}_u) \tag{9}$$

$$-i\mathcal{M} = \frac{-ig^2}{t - m_B^2} + \frac{-ig^2}{u - m_B^2} = -ig^2\left[\frac{1}{t - m_B^2} + \frac{1}{u - m_B^2}\right]$$

Dividing out $-i$:

$$\mathcal{M} = g^2\left[\frac{1}{t - m_B^2} + \frac{1}{u - m_B^2}\right] \tag{10}$$

Now set $m_B = 0$ as the problem specifies:

Total Amplitude ($m_B = 0$)

$$\boxed{\mathcal{M} = g^2\left(\frac{1}{t} + \frac{1}{u}\right) = \frac{g^2(t+u)}{tu}} \tag{11}$$

Compare with Griffiths Eq. 6.55 for $A+A \to B+B$: $\mathcal{M} = -g^2\left[\frac{1}{(p_1-p_3)^2-m_C^2} + \frac{1}{(p_1-p_4)^2-m_C^2}\right]$. **Identical structure**—only the mass in the propagator differs.

5.5 Step 6: Use the Mandelstam Constraint to Simplify $t + u$

The Mandelstam identity (Griffiths Problem 3.25(a)):

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^2$$

With $m_1 = m_2 = m_C = m$ (incoming) and $m_3 = m_4 = m_A = m$ (outgoing):

$$s + t + u = 4m^2 \qquad \Longrightarrow \qquad t + u = 4m^2 - s$$

Substituting:

Amplitude in Mandelstam Variables

$$\boxed{\mathcal{M} = \frac{g^2(4m^2 - s)}{tu}} \tag{12}$$

5.6 Step 7: Square the Amplitude

All particles are spin-0 (scalars), so there is **no spin averaging** and **no trace technology**. We simply take the modulus squared:

$$|\mathcal{M}|^2 = \frac{g^4(4m^2 - s)^2}{t^2\, u^2} \tag{13}$$

Note: $(4m^2 - s)^2 = (s - 4m^2)^2$, so we can write this equivalently as $g^4(s - 4m^2)^2/(t^2u^2)$.

5.7 Step 8: Plug into Part (a)

Insert Eq. (13) into Eq. (6):

$$\frac{d\sigma}{dt} = \frac{S \cdot |\mathcal{M}|^2}{16\pi\, s\,(s-4m^2)} = S \cdot \frac{g^4(s-4m^2)^2}{16\pi\, s\,(s-4m^2)\,t^2\,u^2} \tag{14}$$

One factor of $(s-4m^2)$ cancels:

Final Result: $d\sigma/dt$ for $C + C \to A + A$

$$\boxed{\frac{d\sigma}{dt} = \frac{S\, g^4\,(s-4m^2)}{16\pi\, s\, t^2\, u^2}} \tag{15}$$

where:

- $s + t + u = 4m^2$ (Mandelstam constraint with $m_A = m_C = m$)
- $m_B = 0$ (massless propagating particle)
- g = ABC coupling constant
- S = statistical factor for identical final-state particles

6 Physical Checks

Sanity Checks on the Result

1. **Threshold:** As $s \to 4m^2$ (just enough energy to create two A's), $d\sigma/dt \to 0$. Correct—at threshold the particles have zero CM momentum and can't scatter.
2. **Forward/backward divergence:** $d\sigma/dt \to \infty$ as $t \to 0$ or $u \to 0$. This means $\theta^* \to 0$ or $\theta^* \to \pi$. This is the signature of **massless exchange**—the virtual B mediates a long-range $1/r$ potential, just like the photon in Coulomb/Rutherford scattering. If $m_B \neq 0$, the propagator $1/(t - m_B^2)$ would regulate these divergences.
3. **Identical particle symmetry:** The expression contains $t^2u^2 = (tu)^2$, which is symmetric under $t \leftrightarrow u$. This is required because swapping the two identical A's ($p_3 \leftrightarrow p_4$) exchanges t and u, and the physics must be unchanged. ✓
4. **Dimensions** (natural units): $[g^4] = \text{mass}^4$, $[s\,t^2\,u^2] = \text{mass}^{10}$, $[s - 4m^2] = \text{mass}^2$. So $[d\sigma/dt] = \text{mass}^{-4}$, which is $[\text{length}^2/\text{mass}^2] = [\text{area/energy}^2]$. ✓
5. **Comparison with Griffiths Eq. 6.59:** Griffiths gets $d\sigma/d\Omega \propto 1/(p^2 \sin^2\theta)^2$ for $A + A \to B + B$ with $m_C = 0$. Our result in the CM frame would produce the same angular structure (since $tu \propto p^4 \sin^2\theta$ for equal masses), confirming consistency.

7 Summary: The Complete Calculation Pipeline

Here is the entire logical flow in one place:

Step	Action	Result
1	Start with Thomson Eq. 3.37	$\dfrac{d\sigma}{dt} = \dfrac{\lvert\mathcal{M}\rvert^2}{64\pi\, s\, p_i^{*2}}$
2	Set $m_1 = m_2 = m$ in p_i^{*2}	$p_i^{*2} = \dfrac{s - 4m^2}{4}$
3	Substitute	$\dfrac{d\sigma}{dt} = \dfrac{\lvert\mathcal{M}\rvert^2}{16\pi\, s\,(s - 4m^2)}$
4	Draw t- and u-channel diagrams	Two diagrams, internal B
5	Apply Feynman rules to each	$\mathcal{M}_t = \dfrac{g^2}{t}, \quad \mathcal{M}_u = \dfrac{g^2}{u}$
6	Add amplitudes	$\mathcal{M} = g^2\left(\dfrac{1}{t} + \dfrac{1}{u}\right) = \dfrac{g^2(t+u)}{tu}$
7	Apply $s + t + u = 4m^2$	$\mathcal{M} = \dfrac{g^2(4m^2 - s)}{tu}$
8	Square	$\lvert\mathcal{M}\rvert^2 = \dfrac{g^4(s - 4m^2)^2}{t^2u^2}$
9	Plug into Step 3	$\dfrac{d\sigma}{dt} = \dfrac{S\, g^4(s - 4m^2)}{16\pi\, s\, t^2u^2}$

8 Practice Questions

1. **Why no s-channel?** Draw what an s-channel diagram for $C + C \to A + A$ would look like. Label the particle types at each vertex. Explicitly show which vertex rule is violated.

2. **What if $m_B \neq 0$?** Re-derive $d\sigma/dt$ keeping m_B general. Show that the forward divergence at $t \to 0$ is regulated.

3. **CM-frame version.** Using $t = -(s - 4m^2)(1 - \cos\theta^*)/2$ and $u = -(s - 4m^2)(1 + \cos\theta^*)/2$, convert the result to $d\sigma/d\Omega^*$ and compare with Griffiths Eq. 6.59.

4. **Different process.** Repeat the full calculation for $A + B \to A + B$ (elastic scattering). Which channels are allowed now? [Hint: *all three* channels—s, t, and u—are allowed. Which particle propagates in each?]

5. **Crossing symmetry preview.** The amplitude for $C + C \to A + A$ and for $C + A \to C + A$ are related by crossing ($p_3 \to -p_2'$, $p_2 \to -p_3'$). What substitution on s, t, u relates the two processes?